Roll No. ...............................

# TBT–702(b)

## B. TECH. (BIOTECHNOLOGY) (SEVENTH SEMESTER)

## MID SEMESTER EXAMINATION, Oct., 2022

STEM CELL TECHNOLOGY

**Time : 1½ Hours**

**Maximum Marks : 50**

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Differentiate between Embryonic Stem cells and Adult Stem cells.

10 Marks (CO2 CO3)

OR

(b) Highlight the ethical issues pertaining to stem cell technology.

10 Marks (CO2 CO3)

2. (a) Diagrammatically explain the methodology adopted for the isolation of stem cells. 10 Marks (CO1 CO2)

OR

(b) Explain with the help of diagram, the signaling pathway required for the cell fate determination. 10 Marks (CO1 CO2)

*P. T. O.*

3. (a) "Process of transplanting nuclei from adult cells into oocytes or blastocysts and allowing them to grow and differentiate, producing pluripotent cells." Explain such methodology in detail.

10 Marks (CO2 CO1)

OR

(b) Describe the tests which have been utilized for the identification of stem cells. 10 Marks (CO2 CO1)

4. (a) Explain the causes and medical importance of *in-vitro* fertilization.

10 Marks (CO2 CO1)

OR

(b) Explain the significance of Notch Delta signaling with the help of a diagram. 10 Marks (CO2 & CO1)

5. (a) Discuss the methods adopted for the storage of stem cells. Also highlight the contamination caused during storage. 10 Marks (CO2)

OR

(b) "First system identified for the induction of a programmed DNA rearrangement required for cellular differentiation." Explain such system in detail. 10 Marks (CO2)